

Problem 8.1 A hollow steel shaft is made to replace a solid wrought iron shaft of the same external diameter, the material being 35 percent stronger than the iron. Find what fraction of the outside diameter the internal diameter may be. Also neglecting the couplings, find the percentage saving in weight by this substituting assuming that steel is 2 percent heavier than wrought iron.

Solution:

For the solid shaft, $J_S = \frac{\pi}{32} D^4$

The external diameter of hollow shaft = Diameter of solid shaft = D

For the hollow shaft, $J_H = \frac{\pi}{32} (D^4 - d^4)$

Let τ_s = shear stress intensity for wrought iron shaft

τ_H = shear stress intensity for steel shaft = 1.35 τ_s

Now,

$$T_s = \frac{\pi}{16} D^3 \tau_s$$

$$T_H = \frac{\pi}{16} \left(\frac{D^4 - d^4}{D} \right) \times 1.35 \tau_s$$

The torque transmitted in the two cases will be same

$$\therefore \frac{\pi}{16} D^3 \tau_s = \frac{\pi}{16} \left(\frac{D^4 - d^4}{D} \right) \times 1.35 \tau_s$$

$$\therefore \left(\frac{d}{D} \right)^4 = \frac{0.35}{1.35} \quad \therefore \frac{d}{D} = 0.7135$$

Area of cross – section of the solid wrought iron shaft = $\frac{\pi}{4} D^2$

$$\begin{aligned} \text{Area of cross-section of the hollow steel shaft} &= \frac{\pi}{4} (D^2 - d^2) \\ &= \frac{\pi}{4} D^2 \left(1 - \frac{d^2}{D^2} \right) \end{aligned}$$

$$= \frac{\pi}{4} D^2 (1 - 0.7135^2) = \frac{\pi}{4} D^2 \times 0.49$$

Let ω_S be the weight of unit volume of wrought iron, ω_H be the weight of unit volume of steel = 1.02 ω_S .

$$\therefore \text{Weight of unit length of wrought iron shaft} = \frac{\pi}{4} D^2 \omega_S$$

$$\text{Weight of unit length of hollow steel shaft} = \frac{\pi}{4} D^2 \times 0.49 \times 1.02 \omega_S$$

$$\therefore \frac{\text{Weight of steel shaft}}{\text{Weight of iron shaft}} = \frac{\frac{\pi}{4} D^2 \times 0.49 \times 1.02 \omega_S}{\frac{\pi}{4} D^2 \omega_S} = 0.5$$

Therefore the saving in weight is 50%.

Problem 8.2 Compare the weight of a solid shaft with that of a hollow one to transmit a given power at a given speed with a given maximum shear stress, the inside diameter of the hollow shaft being $\frac{2}{3}$ of the outside diameter.

Solution:

$$\frac{T}{J} = \frac{\tau_s}{R} \quad T \text{ and } \tau_s \text{ being the same} \quad \frac{J}{R} \text{ will be the same for the two shafts.}$$

Let d = Diameter of solid shaft

d_1 = External diameter of hollow shaft

d_2 = Internal diameter of hollow shaft = $\frac{2}{3} d_1$

for solid shaft

$$\frac{J_S}{R_S} = \frac{\pi d^4}{32} \times \frac{2}{d} = \frac{\pi d^3}{16}, \text{ Area } A_s = \frac{\pi}{4} d^2$$

For hollow shaft

$$\text{Area, } A_H = \frac{\pi}{4} \left(d_1^2 - \frac{4}{9} d_1^2 \right) = \frac{5\pi d_1^2}{36}$$

$$J_H = \frac{\pi}{32} \left(d_1^4 - \frac{16}{81} d_1^4 \right) = \frac{65\pi d_1^4}{32 \times 81}$$

$$\frac{J_{PH}}{R_H} = \frac{65\pi d_1^4}{32 \times 81} \times \frac{2}{d_2} = \frac{65\pi d_1^3}{16 \times 81}$$

$$\therefore \frac{\pi d^3}{16} = \frac{65\pi d_1^3}{16 \times 81} \quad \text{or } d_3 = \frac{65}{81} d_1^3 \quad \text{or } d = 0.929 d_1$$

$$\frac{\text{Weight of solid shaft}}{\text{Weight of hollow shaft}} = \frac{A_s}{A_H} = \frac{\frac{\pi}{4} d^2}{\frac{5\pi d_1^2}{36}} = \frac{9}{5} \times \left(\frac{d}{d_1}\right)^2$$

$$= \frac{9}{5} \times (0.929)^2 = 1.553$$

Problem 8.3 Prove that a hollow shaft is always stronger than a solid shaft of the same material, weight and length, when subjected to simple torque.

Solution:

Let d = Internal diameter of hollow shaft

kd = External diameter of hollow shaft so that n is greater than 1

D = Diameter of solid shaft

Since the two shafts are of equal length and weight, their areas of cross-sections are equal.

$$\frac{\pi}{4} d^2 = \frac{\pi}{4} (k^2 d^2 - d^2) \quad \text{or} \quad D^2 = d^2 (k^2 - 1) \quad \therefore D = d \sqrt{k^2 - 1}$$

Torque transmitted by the hollow shaft

$$= \frac{\pi}{16} \times \frac{k^4 d^4 - d^4}{nd} \times \tau_s = \frac{\pi}{16} \times \frac{k^4 - 1}{k} \times d^3 \tau_s \quad \dots\dots(8.20)$$

Torque transmitted by the solid shaft

$$= \frac{\pi}{16} D^3 \tau_s = \frac{\pi}{16} d^3 (k^2 - 1)^{3/2} \tau_s \quad \dots\dots(8.21)$$

The hollow shaft will be stronger than the solid shaft if expression (i) is greater than expression (ii).

$$\frac{k^4 - 1}{k} > (k^2 - 1)^{3/2}$$

$$\text{or} \quad \frac{k^2 + 1}{k} > (k^2 - 1)^{1/2}$$

$$\text{or} \quad (k^2 + 1)^2 > k^2 (k^2 - 1)$$

$$\text{or} \quad k^4 + 2k^2 + 1 > k^4 - k^2$$

$$\text{or} \quad 3k^2 + 1 > 0 \text{ which is true.}$$

Problem 8.4 A steel shaft transmits 105 kW at 150 rpm. If the shaft is 100 mm in diameter, find the torque on the shaft and the maximum shearing stress induced. Find also the twist of the shaft in a length of 6 m. Take $C = 8 \times 10^4 \text{ N/mm}^2$.

Solution: Power transmitted $P = \frac{2\pi NT}{60000}$

$$105 = \frac{2 \times 150 T}{60000} \quad \therefore T = \frac{105 \times 60000}{2\pi \times 150} \text{ Nm} = 6683.73 \text{ Nm}$$

$$T = \tau \frac{\pi d^3}{16}$$

$$\therefore \tau = \frac{16 T}{\pi d^3} = \frac{16 \times 6683.73 \times 10^3}{\pi \times 100^3} = 34.06 \text{ N/mm}^2.$$

Twist of the shaft, $\theta = \frac{T}{J_p C} \cdot \frac{L}{C} = \frac{6683.73 \times 10^3}{\left(\frac{\pi \times 100^4}{32} \right) \cdot 8 \times 10^4} \text{ radian}$

$$= 0.05108 \text{ radian}$$

Problem 8.5 A solid shaft is subjected to a torque of 15 kNm. Find the necessary diameter of the shaft if the allowable shearing stress is 60 N/mm^2 and the allowable twist is 1 degree in a length of 20 diameters of the shaft, Take $C = 7.8 \times 10^4 \text{ N/mm}^2$.

Solution: Shear stress consideration $T = \frac{\tau \pi d^3}{16} \quad \therefore d^3 = \frac{16 \times 15 \times 10^6}{60 \pi} \quad \therefore d = 108.4 \text{ mm}$

Twist consideration $\frac{T}{J} = \frac{C \theta}{L} \quad \therefore \frac{32 T}{\pi d^4} = \frac{C \theta}{L} \quad \therefore d^4 = \frac{32 T L}{\pi C \theta} = \frac{32 \times 15 \times 10^6 \times 20d}{\pi \times 7.8 \times 10^4 \times \left(\frac{\pi}{180} \right)}$

$$\therefore d = 133.13 \text{ mm}$$

$$\therefore \text{Minimum required diameter} = 133.13 \text{ mm}.$$

Problem 8.6 A hollow shaft is to have an outside diameter d and inside diameter $\frac{d}{2}$.

Calculate the minimum value of d if it is to transmit 375 kW at 100 rpm with a working stress of 40 N/mm^2 . Determine the twist in a length equal to 10 times the external diameter. Take $C = 8 \times 10^4 \text{ N/mm}^2$.

Solution: Polar moment of inertia of the shaft section

$$J = \frac{32}{\pi} \left[d^4 - \left(\frac{d}{2} \right)^4 \right] = \frac{32}{\pi} \cdot \frac{\pi}{16} d^4 \text{ mm}^4 = 0.092 d^4 \text{ mm}^4$$

$$Z_p = \frac{J}{\frac{d}{2}} = 0.184 d^3 \text{ mm}^3$$

$$P = \frac{2\pi N T}{60000} \therefore T = \frac{60000 \times 375}{2\pi \times 100} = 35828.02 \text{ Nm}$$

$$T = \tau Z_p = 40 \times 0.184 d^3 = 35828.02 \times 10^3$$

$\therefore$

$$d = 169.5 \text{ mm}$$

$$l = 1695 \text{ mm}$$

$$J_p = \frac{l}{c} \cdot \frac{T}{\theta}$$

$$\theta = \frac{l}{c} \cdot \frac{T}{J_p}$$

$$= \frac{1695}{8 \times 10^4} \times \frac{35828.02 \times 10^3}{0.092 \times (169.5)^4} = 0.01 \text{ radian}$$

Problem 8.7 Find the diameter of the shaft required to transmit 60 kW at 150 r.p.m. if the maximum torque is likely to exceed the mean torque by 20% for a maximum permissible shear stress of 60 N/mm².

Find also the angle of twist for a length of 2.5 meters. Take $C = 8 \times 10^4 \text{ N/mm}^2$.

Solution:

$$P = \frac{2\pi NT}{60000}$$

$$60 = \frac{2\pi \times 150 T}{60 \times 60000}$$

$$T = \frac{60 \times 60000}{2\pi \times 150} = 3819.7 \text{ Nm} = 3819.7 \times 1000 \text{ Nmm.}$$

The torque calculated above is the mean torque

$\therefore$

$$\begin{aligned} \text{Max. torque} &= T_{\max} = 1.20 \times \text{mean torque} \\ &= 1.20 \times 3819.7 \times 1000 = 4583.64 \times 1000 \text{ Nmm} \\ &= \frac{r_{\max}}{32 \times d} = \frac{\pi}{16} \end{aligned}$$

Polar modulus

$$T_{\max} = \tau \times \text{Polar modulus}$$

$\therefore$

$$4583.64 \times 1000 = 60 \times \frac{16}{4583.64 \times 1000 \times 16}$$

$\therefore$

$$d^3 = \frac{60\pi}{4583.64 \times 1000 \times 16}$$

$\therefore$

$$d = 73.02 \text{ mm}$$

$$\frac{T}{J} = \frac{C\theta}{l}$$

$$\therefore \theta = \frac{T}{J} \cdot \frac{l}{C} = \frac{4583.64 \times 1000 \times 32}{\pi \times 73.02^4} \times \frac{2500}{8 \times 10^4}$$

$$= 0.0513 \text{ radian} = 2^{\circ}56'.$$

Problem 8.8 A solid shaft of 200 mm diameter has the same cross-sectional area as that of a hollow shaft of the same material with inside diameter 150 mm. Find the ratio of power transmitted by the two shafts at the same speed.

Solution: Let

D_1 = External diameter of the hollow shaft

D_2 = 150 mm = Internal diameter of the hollow shaft

D = 200 mm = diameter of the solid shaft

$$\frac{\pi}{4} D_1^2 - \frac{\pi}{4} D_2^2 = \frac{\pi}{4} D^2$$

$$\therefore D_1^2 - 150^2 = 200^2$$

$$\therefore D_1 = 250 \text{ mm}$$

Ratio of power transmitted

$$Z_{\text{hollow}} = \frac{16 D_1}{\pi (D_1^4 - D_2^4)} = \frac{\pi (250^4 - 150^4)}{16 \times 250} = \pi \times 8.5 \times 10^5 \text{ mm}^3$$

$$\frac{Z_{\text{solid}}}{P_{\text{hollow}}} = \frac{T_{\text{hollow}}}{T_{\text{solid}}} = \frac{Z_{\text{hollow}}}{Z_{\text{solid}}} = \frac{8.5 \times 10^5 \pi}{5 \times 10 \pi} = 1.7$$

Problem 8.9 A hollow shaft is to transmit 300 kW at 100 rpm. If the shear stress is not to exceed 60 N/mm² and the internal diameter is 0.6 of the external diameter, find the external and internal diameters, assuming the maximum torque is 1.4 times the mean torque.

$$\text{Solution: Power transmitted } P = \frac{2\pi NT_{\text{mean}}}{60000} = 300$$

$$\therefore T_{\text{mean}} = \frac{300 \times 60000}{2\pi \times 100} = 28662.42 \text{ Nm}$$

$$\therefore T_{\text{max}} = 1.4 T_{\text{mean}} = 1.4 \times 28662.42 = 40127.4 \text{ Nm}$$

$$= 40127.4 \times 1000 \text{ Nmm}$$

$$\text{Polar modulus} \quad Z_p = \frac{\pi}{16D} (D^4 - d^4) = \frac{\pi}{16D} [D^4 - 0.6^4 D^4] = 0.1709 D^3$$

$$T_{\max} = \tau Z_p = 40127.4 \times 1000$$

$$\therefore 60 \times 0.1709 D^3 = \frac{40127.4 \times 1000}{1000}$$

$$\therefore D^3 = \frac{40127.4}{60 \times 0.1709}$$

$$\therefore D = 157.58 \text{ mm}$$

$$\therefore d = 0.6 \times 157.58 = 94.55 \text{ mm}$$

Problem 8.10 The shaft shown in Fig. 8.6 rotates at 200 r.p.m. with 30 kW and 20 kW taken off at A and B respectively and 50 kW applied at C. find the maximum shear developed in the shaft and the angle of twist (degree) of the gear A relative to C. Assume $G = 8.5 \times 10^4 \text{ N/mm}^2$.

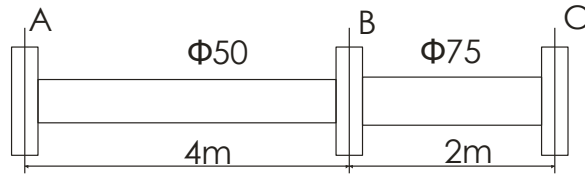


Fig. 8.6

Solution:

Shaft between B and C

Power of the shaft = 50 kW

Let the torque in this part of the shaft be T_{bc}

$$\text{Power} = \frac{2\pi NT}{60 \times 1000}$$

$$50 = \frac{60000}{50 \times 60000} \times T_{bc}$$

$$\therefore T_{bc} = \frac{2\pi \times 200}{50 \times 60000} \text{ Nm} = 2387.33 \text{ Nm} = 2387.33 \times 100 \text{ Nmm}$$

Let τ be the maximum shear stress in this part of the shaft.

$$\tau \cdot \frac{\pi d^3}{16} = T_{bc}$$

$$\tau = \frac{16T_{bc}}{\pi d^3} = \frac{16 \times 2387.33 \times 1000}{\pi \times 75^3} = 28.83 \text{ N/mm}^2.$$

Shaft between B and A

Power of the shaft = 50 – 20 = 30 kW

Let the torque in this part of the shaft be T_{ab}

$$\therefore 30 = \frac{60000}{30 \times 60000}$$

$$T_{ab} = \frac{400\pi}{30 \times 60000} = 1432.4 \text{ Nm}.$$

Let τ be the maximum shear stress in this part of the shaft

$$\therefore \tau \times \frac{\pi d^3}{16} = T_{ab}$$

$$\therefore \tau = \frac{16 T_{ab}}{\pi d^3} = \frac{16 \times 14324 \times 1000}{\pi \times 50^3} = 58.36 \text{ N/mm}^2.$$

Hence the greatest shear stress occurs in the 50 mm. diameter shaft.

$$\therefore \text{Max. shear stress} = 58.36 \text{ N/mm}^2$$

Twist of the shaft

Let θ_{bc} be the twist of the shaft BC.

$$\therefore \theta_{bc} = \frac{T_{bc}}{C} \cdot \frac{L_{bc}}{J}$$

$$\frac{2000}{8.5 \times 10^4} \times \frac{2387.33 \times 1000 \times 32}{\pi \times 75^4} = 0.01809 \text{ radian}$$

Let θ_{ab} be twist of the shaft AB

$$\therefore \theta_{ab} = \frac{T_{ab}}{C} \cdot \frac{L_{ab}}{J}$$

$$= \frac{4000}{8.5 \times 10^4} \times \frac{14326 \times 1000 \times 32}{\pi \times 50^4} = 0.10986 \text{ radian}$$

$\therefore$ Angle of twist of A with respect to C

$$= \theta_{ab} + \theta_{bc} = 0.10986 + 0.01809 = 0.12795 \text{ radian} = 7^\circ 24'$$

Problem 8.11 Two shaft of the same material are subjected to the same torque. If the first shaft is solid circular section and the second shaft is of hollow section whose internal diameter is $\frac{2}{3}$ of the outside diameter, compare the weights of the two shafts.

Solution: The maximum torque a shaft section can safely resist

$$T = \tau \cdot Z_p$$

Where

τ = permissible shear stress

Z_p = polar modulus of shaft section

In order the two shafts may have the same strength to resist the torque, the polar moduli of the shafts must be equal.

Let D be the diameter of the solid shaft section.

$$= \frac{\pi D^3}{16}$$

Let D_1 be the external diameter of the hollow shaft.

$$\therefore \text{Internal diameter} = \frac{2}{3} d_1$$

$\therefore$ Polar modulus of hollow shaft section

$$= \frac{\pi \left(d_1^4 - \frac{16}{81} d_1^4 \right)}{16 d_1} = \frac{65 \pi d_1^3}{81 \cdot 16}$$

Equating the polar moduli of the two shaft sections, we have,

$$\frac{65}{81} \frac{\pi d_1^4}{16} = \frac{\pi d^4}{16}$$

$$\therefore \frac{d_1}{d} = \sqrt[4]{\frac{81}{65}} = 1.075$$

Since the weight of the shaft is the same, the area of the hollow shaft section

$$\begin{aligned} \frac{\text{Weight of solid shaft}}{\text{Area of solid shaft section}} &= \frac{\text{Weight of hollow shaft}}{\text{Area of hollow shaft section}} \\ \frac{\pi d^2}{4} &= \frac{\pi (d_1^2 - \frac{4}{5} d_1^2)}{4} = \frac{5}{9} \left(\frac{d_1}{d} \right)^2 \\ &= \frac{5}{9} (1.075)^2 = 0.642 \end{aligned}$$

Problem 8.12 Fig. 8.7 shows a compound shaft of diameter 60 mm. The part AB 1.5 m long is of Aluminium, while the part BC 1 m long is of Brass. Find the maximum torque that can be applied at the end C if the shear stresses in Aluminium and brass are not to exceed 80 N/mm² and 60 N/mm² respectively and the total angle of twist shall not exceed 1°. Take rigidity moduli for Aluminium and brass as 3 × 10⁴ N/mm² and 3.5 × 10⁴ N/mm².

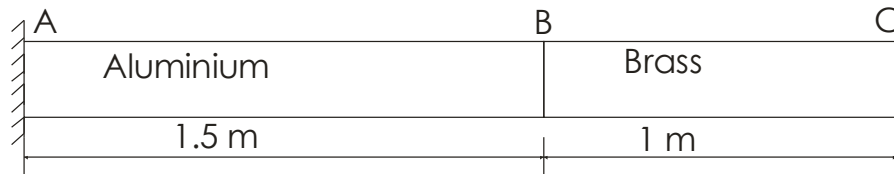


Fig. 8.7

Solution: (i) Torque required to produce a shear stress of 60 N/mm² in Brass

$$T = \frac{\tau J_p}{R} = \tau Z_p = 60 \times \pi \times \frac{60^3}{16} = 2.545 \times 10^6 \text{ Nmm}$$

(ii) Torque required to produce a shear stress of 80 N/mm² in Aluminium

$$T = \tau Z_p = \tau Z_p = 80 \times \frac{\pi \times 60^3}{16} = 3.393 \times 10^6 \text{ Nmm}$$

(iii) Torque required to produce a twist of 1°.

$$Q = \frac{T_b}{J C_b} + \frac{T_a}{J C_a} = \frac{T}{J} \left(\frac{1}{C_b} + \frac{1}{C_a} \right) = \frac{T}{180}$$

$$\frac{32T}{\pi \times 60^4} \left[\frac{1000}{3.5 \times 10^4} + \frac{1500}{3 \times 10^4} \right] = \frac{\pi}{180} \quad \therefore T = 0.2823 \times 10^6 \text{ N/mm}$$

$\therefore$ Maximum allowable torque = least of the value of T obtained above = 0.2823×10^6 Nmm

Problem 8.13 A hollow steel shaft 240 mm external and 160 mm internal diameter is to be replaced by a solid alloy shaft. If both the shafts should have the same polar modulus, find the diameter of the latter and the ratio of the torsional rigidities. Take C for steel = $2.5 \times 10^4 \text{ N/mm}^2$ for alloy. If alternatively, the two shafts should have the same torsional rigidity, find the ratio of their polar moduli.

Solution: Case (i) When the polar moduli of the two shafts are equal.

i.e. $Z_{\text{steel}} = Z_{\text{alloy}}$

$$\begin{aligned} \text{Polar modulus of the steel shaft} &= Z_{\text{steel}} = \frac{\pi}{16} d_1^3 [d_1^4 - d_2^4] \\ &= \frac{\pi}{16 \times 240} (240^4 - 160^4) \text{ mm}^3 = \frac{2080}{3} \times 10^3 \pi \text{ mm}^3 \end{aligned}$$

Let the diameter of the solid alloy shaft be d mm

$$\therefore \text{Polar modulus of the alloy shaft} = Z_{\text{alloy}} = \frac{\pi d^3}{16}$$

$\therefore$ Equating the polar moduli of the two shafts, we get,

$$\begin{aligned} \frac{\pi d^3}{16} &= \frac{2080}{3} \times 10^3 \pi \\ \therefore d^3 &= \frac{2080 \times 16}{3} \times 10^3 \end{aligned}$$

$$\therefore \text{Ratio of torsional rigidities} = \frac{\text{Torsional rigidity of steel shaft}}{\text{Torsional rigidity of alloy shaft}} = \frac{C_s I_s}{C_a I_a}$$

But since the polar moduli are equal

$$\begin{aligned} \frac{J_s}{r_s} &= \frac{J_a}{r_a} \\ \frac{J_s}{J_a} &= \frac{r_s}{r_a} \end{aligned}$$

$$\therefore \text{Ratio of torsional rigidities} = \frac{C_s}{C_a} \cdot \frac{r_s}{r_a} = 2.5 \times \frac{120}{111.5} = 2.63$$

Case (ii). When the torsional rigidities of the two shafts are equal

$$\frac{\pi}{32} C_s J_s = C_a J_a \quad \frac{\pi}{32} D^4$$

$$D_4 = \frac{C_s}{C_a} (240^4 - 160^4) = 2.5 (240^4 - 160^4)$$

$$D = \frac{25.63 J_a}{Z_a} = \frac{25.63 J_a}{J_a r_s}$$

∴ Ratio of polar moduli =

Since the torsional rigidities are equal

$$\frac{C_a J_s}{J_a} = \frac{C_s J_a}{C_a} \frac{1}{2.5}$$

$$\frac{Z_s}{Z_a} = \frac{1}{2.5} \times \frac{142.815}{120} = 0.47605.$$

Problem 8.14 The stepped steel shaft shown in Fig. 8.8 is subjected to a torque T at the free end and a torque 2T in the opposite direction at the junction of the two sizes. What is the total angle of twist at the free end if the maximum shear stress in the shaft is limited to 80 N/mm²? Assume the modulus of rigidity to be 8.4 × 10⁴ N/mm².

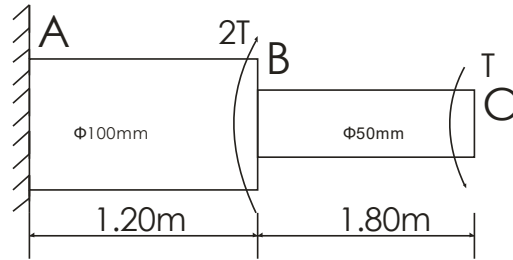


Fig. 8.8

Solution: Torque on the shaft BC = T

∴ Torque on the shaft AB = T - 2T = - T

Hence the two shafts are subjected to a torque of same magnitude but of opposite sense.

Hence the shaft BC which is of smaller diameter will be subjected to a greater shear stress than the shaft AB.

Equating the torsional strength of the shaft BC to the external torque, we have,

$$\tau \frac{16}{\pi} = T$$

$$\therefore T = \frac{80 \times \pi \times 50^3}{16} = 1963497 \text{ Nmm}$$

Let the twist of the shaft BC be θ_{bc} radians,

$$\therefore \frac{T}{J_{bc}} = \frac{1963497}{\left(\frac{\pi \times 50^4}{32} \right)} \times \frac{1800}{8.4 \times 10^4}$$

$$\therefore \theta_{bc} = \left(\frac{\pi \times 50^4}{32} \right) \times \frac{1963497}{8.4 \times 10^4} = 0.0686 \text{ radian}$$

Similarly the twist θ_{ab} of shaft AB is given by,

$$\theta_{ab} = \left(\frac{\pi \times 100^4}{32} \right) \times \frac{1200}{8.4 \times 10^4} = 0.0043 \text{ radian}$$

Since the directions of twist θ_{bc} and θ_{ab} are opposite to each other.

$$\begin{aligned} \text{Net angle of twist of the free end} &= \theta_{bc} - \theta_{ab} = 0.0686 - 0.0043 = 0.0643 \text{ radian} \\ &= 3^\circ 30' \end{aligned}$$

Problem 8.15 A solid shaft 7 meters long is securely fixed at each end. A torque of 91 Nm is applied to the shaft at a section 3 meters from one end. Find the fixing torques set up at the ends of the shaft. If the shaft is 35 mm diameter find the maximum shear stress in the two portions. Find also the angle of twist for the section where the torque is applied. Take $C = 8.4 \times 10^4 \text{ N/mm}^2$.

Solution: Fig. shows the shaft AB. Let the fixing couple at the ends A and B be T_a and T_b respectively.

$$\therefore T_a + T_b = T \quad \dots (8.22)$$

Since the fixing couple must oppose the applied couple,

Further the ends being fixed the angle of twist in the length AC of the shaft must be equal to the angle of the twist in the length CB.

The twist θ for a length l is given by

$$\theta = \frac{TL}{CJ}$$

Since C and I_p have the same value for the two portions

We have $T_a \cdot AC = T_b \cdot CB$

$$\therefore T_a \cdot (3) = T_b \cdot (4)$$

$$\therefore T_a = 1.33 T_b$$

Substituting in equation (8.22), we get

$$1.33 T_b + T_b = 91$$

$$\therefore T_b = \frac{91}{2.33} = 39.1 \text{ Nm}$$

$$\therefore T_a = 91 - 39.1 = 51.9 \text{ Nm}$$

$$\text{Polar modulus} = Z_p = \frac{16}{16} = \frac{16}{16} \text{ mm}^3$$

$$\begin{aligned} \text{Maximum shear stress in the portion AC} &= \frac{T_{AC}}{Z_p} = \frac{51.9 \times 1000 \times 16}{\pi \times 35^3} \\ \tau_{ac} &= \frac{51.9 \times 1000 \times 16}{\pi \times 35^3} = 6.163 \text{ N/mm}^2. \end{aligned}$$

$$\begin{aligned} \text{Maximum shear stress in the portion CB} &= \frac{T_{CB}}{Z_p} = \frac{39.1 \times 1000 \times 16}{\pi \times 35^3} \\ \tau_{cb} &= \frac{39.1 \times 1000 \times 16}{\pi \times 35^3} = 4.65 \text{ N/mm}^2 \end{aligned}$$

The angle of twist can be determined considering either the part AB or CB

$$\theta = \frac{Tl}{CJ} = \frac{T_a \cdot AC}{CJ}$$

$$= \frac{51.9 \times 1000 \times 3000 \times 32}{8.4 \times 10^4 \times \pi \times 35^4} = 0.01238 \text{ radian} = 0^\circ 44' 34''$$

Problem 8.16 A solid steel shaft 50 mm in diameter is fixed rigidly and concentrically inside a bronze sleeve 75 mm external diameter. Find the angle of twist in a 2 meter length of the composite shaft under the action of a torque of 1000 N-m. Modulus of rigidity of steel is $8 \times 10^4 \text{ N/mm}^2$ and that of bronze is $4 \times 10^4 \text{ N/mm}^2$.

Solution:

$$J_s = \frac{\pi}{32} \times 50^4 = 613592 \text{ mm}^4.$$

$$J_b = \frac{\pi}{32} (75^4 - 50^4) = 2492719 \text{ mm}^4$$

$$T_s = \frac{l}{C_b I_b \theta}$$

and

$$T_b = \frac{l}{C_s I_s \theta}$$

The two shafts are arranged in parallel

$$\therefore \text{The total torque } T = T_s + T_b = (C_s I_s + C_b I_b) \frac{\theta}{l}$$

$\therefore$

$$\theta = \frac{Tl}{C_s I_s + C_b I_b}$$

$$= \frac{(1000 \times 10^3)(2 \times 10^3)}{(8 \times 10^4 \times 613592) + (4 \times 10^4 \times 2492719)}$$

$$= 0.01344 \text{ radian} = 48' 23''$$

Problem 8.17 A composite shaft consists of a steel rod 60 mm diameter surrounded by a closely fitting tube of brass fixed to it. Find the outside diameter of tube so that when a torque is applied to the composite shaft, it will be shared equally by the two materials. Take C for steel = $8 \times 10^4 \text{ N/mm}^2$ and C for brass = $4 \times 10^4 \text{ N/mm}^2$.

Solution: Let the twist of each shaft be θ

$$\text{Polar moment of inertia of steel shaft} = J_s = \frac{\pi \times 60^4}{32} \text{ mm}^4$$

$$\text{Polar moment of inertia of the brass shaft } J_b = \frac{\pi}{32} (D^4 - 60^4) \text{ mm}^4$$

If T_s and T_b be the torques in the steel and brass shafts,
We have

$$\frac{T_s}{J_s} = \frac{C_s \theta}{l}$$

$$\frac{T_b}{J_b} = \frac{C_b \theta}{l}$$

and

But $\frac{T_b}{J_s} = \frac{T_s}{C_b} = \frac{8}{4} = 2$

$\therefore$

$\therefore \frac{\pi}{32} (D^4 - d^4) = 2 \times \frac{\pi}{32} d^4$

$\therefore$

$\therefore D^4 - d^4 = 2 \times d^4$

$\therefore$

$\therefore D^4 = 3 \times d^4 = 3 \times 60^4$

$\therefore$

$\therefore D = 78.96 \text{ mm}$

$$T_s = T_b = \frac{5000 \times 1000}{2} = 5000 \text{ Nm} = 5000 \times 1000 \text{ Nmm}$$

For the steel shaft, maximum shear stress $\tau_s = \frac{T_s}{Z_s} = \frac{5000 \times 1000 \times 16}{\pi \times 60^3}$

$$= \tau_s = \frac{5000 \times 1000 \times 16}{\pi \times 60^3} = 117.9 \text{ N/mm}^2$$

For the brass shaft, maximum shear stress $\tau_b = \frac{T_b}{Z_b} = \frac{5000 \times 1000 \times 16D}{\pi(D^4 - d^4)}$

$$= \tau_b = \frac{5000 \times 1000 \times 16D}{\pi(D^4 - d^4)}$$

Since $D^4 - d^4 = 2d^4$

$$q_b = \frac{5000 \times 1000 \times 16 \times 78.96}{\pi \times 2 \times 60^4} \text{ N/mm}^2 = 77.6 \text{ N/mm}^2$$

The common angle of twist $\theta = \frac{5000 \times 1000 \times 4000 \times 32}{8 \times 10^4 \times \pi \times 60^4} \text{ radian}$

$$= 0.1916 \text{ radian} = 11^\circ 17'$$

Problem 8.18 A composite shaft 7.50 m long consists of a steel shaft 250 mm in diameter surrounded by a closely fitting 30 mm thick bronze tube. If the shear stress in the shaft shall not exceed 20 N/mm² find the maximum power transmitted by the shaft at 250 rpm. Take $C_s = 8.4 \times 10^4 \text{ N/mm}^2$ and $C_b = 4.2 \times 10^4 \text{ N/mm}^2$.

Solution: Polar moment of inertia of the steel component, $J_s = \frac{\pi}{32} \times 250^4 \text{ mm}^4$

Polar moment of inertia of the bronze component,

$$J_b = \frac{\pi}{32} (310^4 - 250^4) = \frac{\pi}{32} \times 5.32896 \times 10^8 \text{ mm}^4$$

Let T_s and T_b the torques transmitted to the steel and bronze components. The angle of twist is the same for the two shafts, $\theta = \frac{T_s}{J_s} \cdot \frac{l}{C_s} = \frac{T_b}{J_b} \cdot \frac{l}{C_b}$

$$\therefore \frac{T_s}{T_b} = \frac{J_s}{J_b} \cdot \frac{C_s}{C_b} = \frac{\frac{\pi}{32} \times 250^4}{\frac{\pi}{32} \times 5.32896 \times 10^9} \cdot \frac{8.4 \times 10^4}{4.2 \times 10^4} = 1.466 \quad \therefore T_s = 1.466 T_b$$

When the maximum shear stress in the steel component reaches 15 N/mm^2 ,

$$T_s = \tau_s \cdot \frac{16}{61.36 \times 10^6} = 20 \times \frac{16}{61.36 \times 10^6} = 61.36 \times 10^6 \text{ Nmm}^2,$$

$$\therefore T_b = \frac{61.36 \times 10^6}{1.466} = 41.86 \times 10^6 \text{ Nmm}$$

Total torque on the shaft,

$$T = T_s + T_b = (61.36 + 41.86) \times 10^6 \text{ Nmm} = 103.22 \times 10^6 \text{ Nmm}$$

$$\therefore \text{Power transmitted, } P = \frac{2\pi NT}{60000} = \frac{2\pi \times 250 \times 103.22 \times 10^6}{60000} = 2700.9 \text{ kW}$$

Problem 8.19 A solid alloy shaft 50mm diameter is to be coupled in series with a hollow steel shaft of the same external diameter. Find the internal diameter of the steel shaft if the angle of twist per unit length of the steel shaft is to be 75% of that of the alloy shaft. Determine the speed at which the shafts are to be driven to transmit 18.74 kW if the limits of shearing stress are to be 56 N/mm^2 and 80 N/mm^2 in alloy and steel respectively. Take $C_{\text{steel}} = 2 C_{\text{alloy}}$.

Solution: Angle of twist per unit length of a shaft is given by

$$\frac{\theta}{l} = \frac{T}{CJ_p}$$

$$\therefore \frac{T}{C_s J_s} = 0.75 \frac{T}{C_a J_a}$$

$$\therefore \frac{J_a}{J_s} = 0.75 \frac{C_s}{C_a}$$

$$\therefore \frac{\frac{\pi}{32} \times 50^4}{\frac{\pi}{32} (50^4 - d^4)} = 0.75 \times 2$$

$$\therefore 50^4 - d^4 = \frac{50^4}{0.57 \times 2} = 5482456.14$$

$$\therefore d^4 = 50^4 - 5482456.14 = 767543.86$$

$$\therefore d = 29.6 \text{ mm}$$

Also for a shaft, we know the relation

$$\frac{\tau_s}{r} = \frac{\theta}{l}$$

$$\therefore \frac{\tau_{steel}}{r_{steel}} = C_s \left(\frac{\theta}{l} \right)_{steel} \quad \text{and} \quad \frac{\tau_{alloy}}{r_{alloy}} = C_a \left(\frac{\theta}{l} \right)_{alloy}$$

$$\frac{\tau_{steel}}{\tau_{alloy}} = \frac{C_s}{C_a} \cdot \frac{r_s}{r_a} \times 0.75 = 2 \times 0.75 = 1.50$$

$$\therefore \tau_{steel} = 1.50 \tau_{alloy}$$

When $\tau_{steel} = 80 \text{ N/mm}^2$

$$\tau_{alloy} = \frac{80}{1.5} = 53.33 \text{ N/mm}^2$$

(This is less than the permissible stress of 56 N/mm²)

Now the torque can be determined by considering any one of the shafts.

$$T = \tau_{alloy} \frac{\pi D^3}{16} = \frac{53.33 \times \pi \times 50^3}{16} \text{ Nmm}$$

$$= 1308251.6 \text{ Nmm}$$

$$\therefore \text{Power transmitted} = P = \frac{60000}{2\pi T} = \frac{60000 \times 18.75}{2\pi \times 1308.25}$$

$$\therefore N = 136.9 \text{ rpm.}$$

Problem 8.20 A solid alloy shaft of 50 mm diameter is to be coupled in series with a hollow steel shaft of the same external diameter. Find the internal diameter of the steel shaft if the angle of twist per unit length is to be 75% of that of the alloy shaft. Determine the speed at which the shafts are to be driven to transmit 200 kW if the limits of shearing are to be 50 N/mm² and 70 N/mm² in alloy and steel respectively. Take $C_s = 22C_a$.

Solution: Angle of twist per unit length of a shaft = $\frac{\theta}{l} = \frac{T}{J C}$

Let suffices a and s refer to alloy shaft and steel shaft respectively

As per data,

$$\frac{J_s C_s}{J_a C_a} = \frac{4}{3} \cdot \frac{1}{2.2} = \frac{4}{6.6}$$

$$\therefore I_s = \frac{4}{3} \cdot \frac{C_a}{C_s} J_a = \frac{4}{6.6} J_a$$

$$\frac{\pi}{32} (50^4 - d^4) = \frac{4}{6.6} \cdot \frac{\pi \times 50^4}{32}$$

$$\therefore 50^4 - d^4 = \frac{4}{6.6} \cdot 50^4$$

$$d^4 = 50^4 \left(1 - \frac{4}{6.6} \right)$$

$$\therefore d = 39.61 \text{ mm}$$

Let τ_a = maximum shearing stress in the alloy shaft

τ_s = maximum shearing stress in the steel shaft
 since both shafts have the same maximum radius.

$\frac{\tau_a}{C_s \theta_s} = \frac{\tau_s}{C_s \theta_s}$ and θ_s being the angle of twist per unit length for the alloy and steel shafts.

But $\theta_a = \frac{4}{3}$

$$\therefore \frac{\tau_a}{\tau_s} = \frac{1}{2.2} \cdot \frac{4}{3} = \frac{4}{6.6} = \frac{2}{3.3} \quad \therefore \tau_s = 1.65 \tau_a$$

If $\tau_a = 50 \text{ N/mm}^2$, then $\tau_s = 1.65 \times 50 = 82.5 \text{ N/mm}^2$ which is greater than 70 N/mm^2 .

$\therefore$ Let $\tau_s = 70 \text{ N/mm}^2 \therefore \tau_a = \frac{70}{1.65} = 42.42 \text{ N/mm}^2$.

Consider say the alloy shaft

$$\text{Torque } T_a = \tau_a \frac{\pi d^3}{16} = 42.42 \times \frac{\pi \times 50^3}{16}$$

$$= 1041143.4 \text{ Nmm} = 1041.1434 \text{ Nm}$$

Power transmitted

$$P = \frac{60000}{2\pi T} = \frac{60000 \times 200}{2\pi \times 1041.1434} = 1834 \text{ rpm.}$$

Exercise

8.1 Define torsional rigidity polar moment of inertia.

8.2 Derive the relation for a circular shaft when subjected to torsion mentioning the assumptions made.

8.3 Find an expression for the torque transmitted by a hollow circular shaft.

8.4 What do you mean by 'strength of a shaft'?

8.5 Find an expression for the strain energy stored in a shaft due to torsion.

8.6 A solid shaft of 250 mm diameter has the same cross sectional area as the hollow shaft of the same material with inside diameter of 200 mm. Find the ratio of the power transmitted by the two shafts for the same angular velocity.

(Ans. 1.78)

8.7 A solid shaft of mild steel 240 mm diameter is to be replaced by a hollow shaft of alloy steel for which allowable shear stress is 22 percent greater. The power to be transmitted is to be increased by 20 percent and the speed of the rotation increased by 5 percent. Determine the maximum internal diameter of the hollow shaft.

(Ans. 120 mm)

8.8 A solid alloy shaft of 50 mm diameter is to be coupled in series with a hollow steel shaft of the same external diameter. Find the internal diameter of the steel shaft if the angle of twist per unit length is to be 75 percent of that of the alloy shaft. Determine the speed at which the shafts are to be driven to transmit 185 kW if the limits of shearing stress are to be 55 and 85 MPa in the alloy and steel respectively. $C_{\text{steel}} = 2.2 C_{\text{alloy}}$.

(Ans. 1485 rpm)

8.9 Prove that a hollow shaft is always stronger than a solid shaft of the same material, weight and length when subjected to simple torque.

8.10 A solid steel shaft is required to transmit a maximum torque of 23 kN-m. if the shear stress is not to exceed 75 MPa and the angle of twist should be limited to 0.17° find the diameter of the shaft. Take $C = 84 \text{ GPa}$.

(Ans. 175 mm)